STATISTICS · EXPLORING DATA

Residuals & Influence Lab

Explore & Reflect

Name _____ Date _____ Period _____

Open **Residuals & Influence Lab** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. If r = 0.98, must a straight line be a good model for the data? Explain.

2. If study time changes from hours to minutes, which change: the slope, r, r^2?

2 Explore use the tool

1. Study data: record the LSRL, r, r^2 and s.

2. Interpret the slope and r^2 in context.

3. Age and height data: record r and describe the residual plot. Is a line appropriate?

4. Use the extra point to find (a) an outlier that is not influential, (b) an influential point and (c) a high-leverage point that is not influential. Record coordinates.

3 Record use the tool and fill in the table

One extra point added to the study data

Thresholds: outlier |residual| > 2s; high leverage x more than 2 SDs from $\bar{x}$; influential slope change ≥ 50%.

Extra point	Residual	Slope change	Outlier?	High leverage?	Influential?
(4, 30)					
(12, 0)					
(12, 96)					
(5, 72)					

4 Explain what did you discover?

1. A residual = _____ – _____.
2. A curved pattern in the residual plot means _____.
3. r^2 is the fraction of the variation in y that is _____.
4. A point is influential if removing it _____.
5. Why can a point with high leverage fail to be influential? Use an example from the lab.

5 Apply a new situation, no tool

1. $\hat{y} = 15 + 2.4x$ predicts quiz score from hours of study. A student studied 5 hours and scored 31. Find and interpret the residual.

2. For a scatterplot, $r = -0.8$. What fraction of the variation in y does the linear model account for?

3. Use the study model $\hat{y} = 51.97 + 3.706x$ to predict the score for 30 hours of study. Is the prediction trustworthy?